

Landscaping Bitcoin Core

@stratospher on 28/11/2024

Bitcoin Core

https://github.com/bitcoin/bitcoin

What is Bitcoin Core?

Bitcoin Core connects to the Bitcoin peer-to-peer network to download and fully validate blocks and transactions. It also includes a wallet and graphical user interface, which can be optionally built.

Erlay

https://github.com/bitcoin/bitcoin/issues/30249

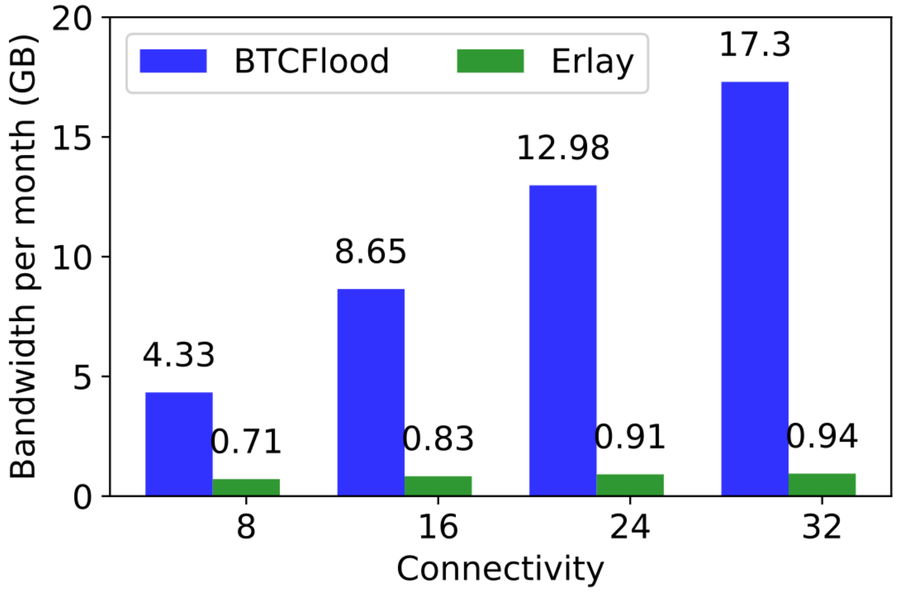


Figure 10: Average bandwidth one Bitcoin node spends per month to announce transactions.

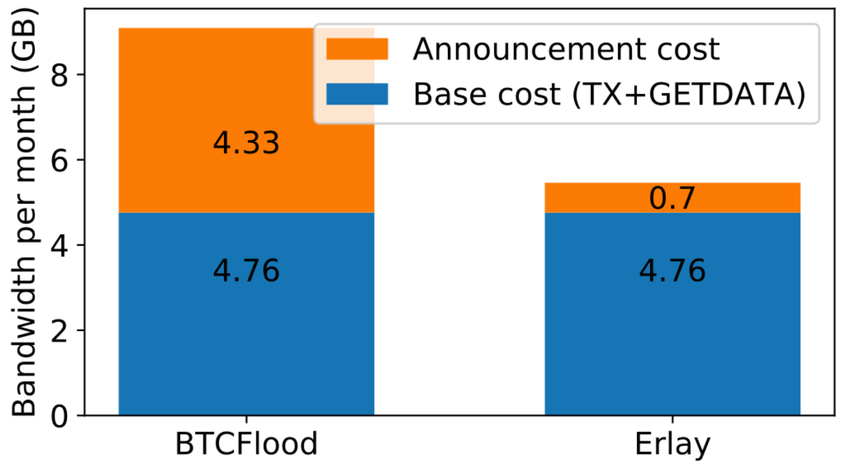
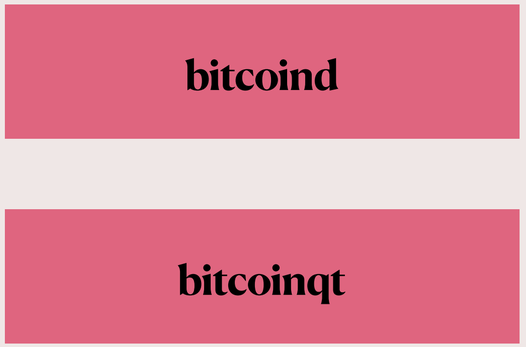


Figure 11: Average bandwidth cost of fully relaying transactions during 1 month for a Bitcoin node with outbound connectivity of 8.

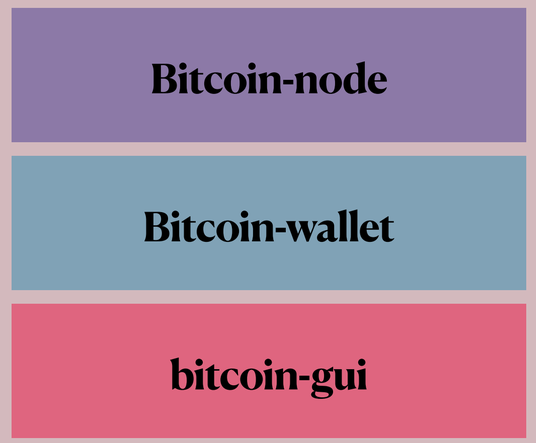
Multiprocess

https://github.com/bitcoin/bitcoin/issues/28722

- Current monolithic architecture

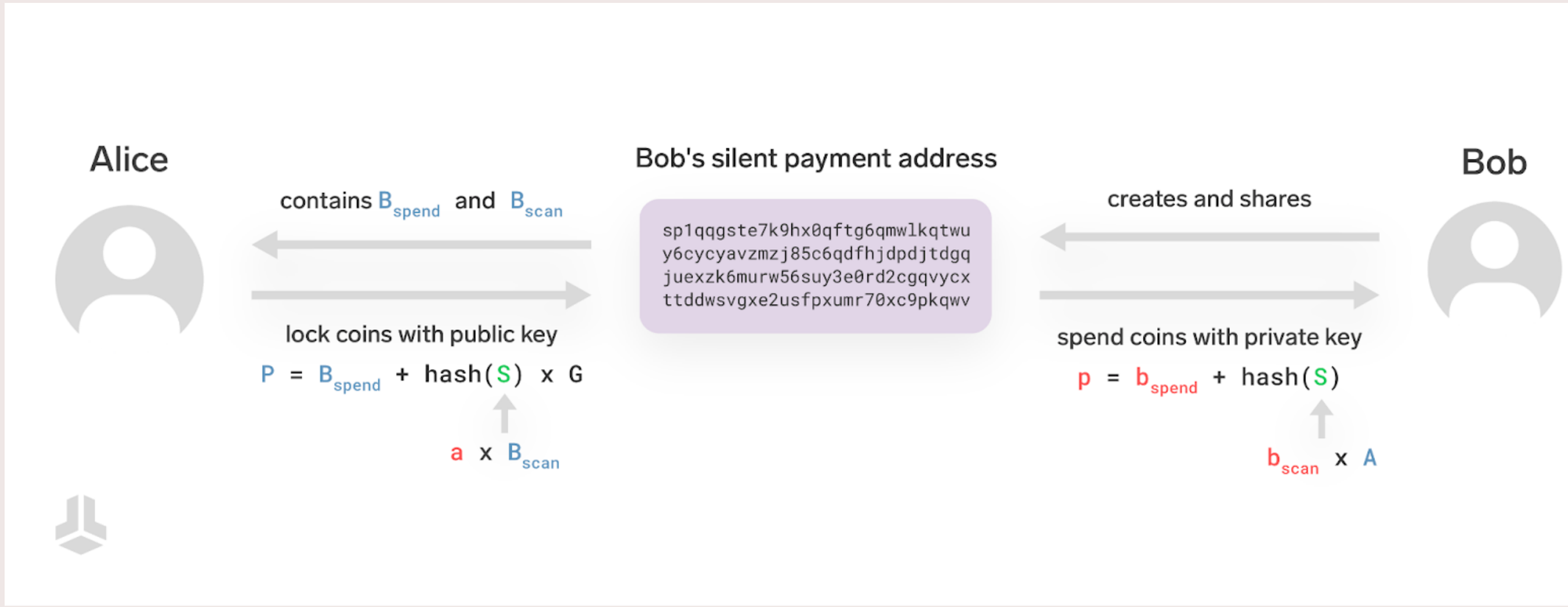


- Modular architecture with multiprocess



BIP 352

https://github.com/bitcoin/bitcoin/issues/28536



Stratum V2

https://github.com/bitcoin/bitcoin/issues/31098

- Include it in core or multiprocess approach?

Fuzzing

- net_processing - https://github.com/bitcoin/bitcoin/issues/27502
- wallet - https://github.com/bitcoin/bitcoin/issues/29901
- https://github.com/brunoerg/bitcoinfuzz : Differential Fuzzing of Bitcoin implementations and libraries

Benchmarking

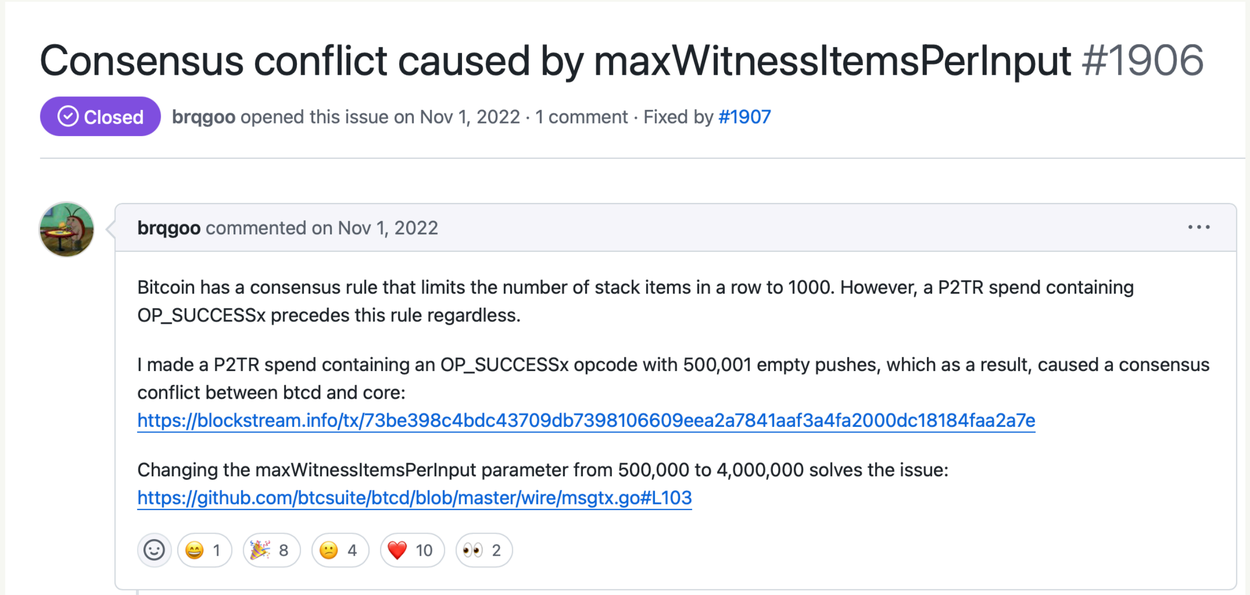
https://btctranscripts.com/bitcoin-core-dev-tech/2024-10/leveldb

- leveldb benchmarking
- Resurrect nightly benchmarking sync from genesis/snapshot to catch regression faster

libbitcoinkernel

https://github.com/bitcoin/bitcoin/issues/27587

- Modularise consensus critical code from the rest
- Other implementations can use libbitcoinkernel as their own internal consensus engine



Consensus conflict caused by maxWitnessItemsPerInput #1906

Closed **brqgoo** opened this issue on Nov 1, 2022 · 1 comment · Fixed by #1907

brqgoo commented on Nov 1, 2022

Bitcoin has a consensus rule that limits the number of stack items in a row to 1000. However, a P2TR spend containing OP_SUCCESSx precedes this rule regardless.

I made a P2TR spend containing an OP_SUCCESSx opcode with 500,001 empty pushes, which as a result, caused a consensus conflict between btcd and core:
https://blockstream.info/tx/73be398c4bdc43709db7398106609eea2a7841aaf3a4fa2000dc18184faa2a7e

Changing the maxWitnessItemsPerInput parameter from 500,000 to 4,000,000 solves the issue:
https://github.com/btcsuite/btcd/blob/master/wire/msgtx.go#L103

1 8 4 10 2

Thank you!

More : Cluster mempool, package relay, legacy wallet removal etc...